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🐙 GitHub Profile

🌐 LinkedIn Profile

EDUCATION

•Bachelor of Computer Engineering

Institut Teknologi Sepuluh Nopember, Surabaya

2022-Present

GPA: 3.64

EXPERIENCE

•Research Intern – Mobile Developer & Computer Vision Engineer

Bandung

Feb - July 2026

Badan Riset dan Inovasi Nasional (BRIN)

a. Full-Stack & Application Development

- Built a cross-platform Flutter app (Mobile, Tablet, Desktop) to calculate NPK nutrient requirements for vegetable crops, deployed to multiple screen sizes.
- Engineered a RESTful API backend with Laravel 12 and PostgreSQL; implemented secure token-based authentication using Laravel Sanctum.
- Designed an intuitive, responsive UI in Flutter, ensuring a seamless experience across all device form factors.

b. AI & Computer Vision Systems

- Developed a visual awareness system leveraging camera input and deep learning models to extract contextual environmental information, including user attributes (e.g., clothing) and surrounding objects.
- Integrated computer vision outputs with a Large Language Model (LLM) to generate enriched contextual understanding, enabling interpretation of user activities such as walking, cycling, motorcycling, and driving.
- Designed and developed a specialized module for a processing pipeline, integrating object detection, human action recognition, and semantic reasoning to enhance situational awareness.

•IoT Developer – Smart Wheelchair Development

Surabaya

Aug 2024 - Present

Multimedia and IoT Laboratory (MIOT)

- Maintained and enhanced a smart wheelchair system utilizing deep learningbased computer vision for handsfree navigation and control.
- Integrated realtime eyegaze and facial gesture recognition using Python and OpenCV, improving system responsiveness and user accessibility.
- Collaborated with faculty and student teams to refine model architectures, optimize inference speed, and troubleshoot hardwaresoftware integration issues.
- Developed custom microcontroller firmware (C/C++) for reliable motor actuation and sensor feedback, ensuring safe and accurate wheelchair movement.
- Documented system upgrades, conducted user testing, and presented results in lab meetings and academic showcases.

•Research Assistant – Multimedia and IoT Lab (B300)

Surabaya

April - June 2025

Multimedia and IoT Laboratory (MIOT)

- Conducted research on deploying deep learning models on edge devices using NVIDIA Jetson Nano, focusing on computer vision tasks.
- Developed and optimized object detection pipelines using Python and C++ with MobileNet-SSD, YOLOv5n, and YOLOv11s architectures.
- Explored model conversion and acceleration techniques with ONNX and NVIDIA TensorRT for real-time inference.
- Integrated and evaluated models within the NVIDIA DeepStream SDK for efficient video analytics and deployment.
- Benchmarked performance and resource utilization to optimize inference speed and accuracy in constrained environments.

•Lab Teaching Assistant – Basic Programming

Aug - Nov 2025

Surabaya

Multimedia and IoT Laboratory (MIOT)

- Developed and maintained lab materials, including coding exercises and troubleshooting guides during practicum, while also being responsible for deploying and maintaining the DMOJ online judge website used in coursework.
- Guided students through practical assignments in C/C++, providing real-time support and technical explanations.
- Evaluated student submissions, provided constructive feedback, and contributed to grading rubrics to ensure fair and consistent assessment.
- Organized and led review sessions before exams, addressing common misconceptions and reinforcing key programming concepts.

•Workshop Assistant – International Workshop on Deep Learning

24 - 29 Nov 2025

Surabaya

Institut Teknologi Sepuluh Nopember

- Assisted in the execution of a 5-day international workshop focused on machine learning and deep learning applications for assistive technology.
- Provided hands-on support to participants by helping resolve technical issues and learning difficulties during workshop sessions.
- Supported professors and speakers in running the event, including administrative coordination and on-site logistical tasks.
- Contributed to creating an inclusive and supportive learning environment for students from Malaysia, Vietnam, the Philippines, Ethiopia, Pakistan, and Indonesia.

PROJECTS

•Sappy: Multi-Role Livestock Management App

Flutter mobile app for farmers, admins, and doctors with NFC, localization, and data visualization.

- Provides role-based access and dashboards for farmers, administrators, and doctors, all in a unified mobile app.
- Supports English and Indonesian languages using built-in localization.
- Implements state management via Provider and enables NFC functionality for secure identification and tracking.
- Includes data visualization, authentication, and environmental configuration using modern Flutter ecosystem packages.
- Technology Used: Flutter, Dart, Provider, NFC, fl_chart, carousel_slider, custom fonts/assets.

•Comic Reading Web Application

Responsive comic platform (manga/manhua/manhwa) with auth, reading history, and chapter management.

- Developed a responsive comic reading platform supporting manga, manhua, and manhwa with smooth page navigation and optimized image rendering.
- Implemented core reading features including chapter management, reading history, authentication, and adaptive layouts for multiple devices.
- Technology Used: React (frontend), Laravel (backend/API), PostgreSQL (database), with modular component architecture and optimized asset handling.

•Smart Wheelchair with Eye Movement Control

Computer vision-driven wheelchair controllable via eye gaze – 4th place in GEMASTIK XVII.

- Developed a computer vision system for real-time eye gaze detection and facial gesture recognition, enabling hands-free control of a motorized wheelchair.
- Integrated eye-tracking algorithms with embedded hardware (Arduino/ESP32) and motor controllers to provide responsive and reliable wheelchair navigation.
- Selected as a National Finalist in GEMASTIK XVII, the premier national competition for technology and innovation among Indonesian university students.
- Technology Used: Python, OpenCV, MediaPipe, Arduino/ESP32, motor control systems.

TECHNICAL SKILLS

Languages: Dart, Python, JavaScript, C/C++, HTML+CSS

Mobile: Flutter (cross-platform), Provider, NFC, responsive UI design

Web & Backend: React.js, Laravel, Node.js, PostgreSQL, REST APIs

Tools: Git, GitHub, VS Code, Figma

Soft Skills: Problem Solving, Fast Learning, Teamwork, Adaptability